GGG-W-657C
February 12, 1976
SUPERSEDING
Int. Fed. Spec. GGG-W-006578(GSA-FSS)
October 5, 1970

FEDERAL SPECIFICATION

WRENCH, SOCKET (SPIN TYPE, SCREWDRIVER GRIP)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers socket wrenches with an integral socket and shaft which utilize a screwdriver type grip.

1.1.1 Federal specification coverage. Federal Specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types and classes. The spin type socket wrenches covered by this specification shall be of the following types and classes when applicable, as specified (see 6.1):

Type I - Conventional.
Type II - Pocket.
Type III - Miniature.
Type IV - Stubby.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

VV-B-680 - Brake Fluid, Automotive.
PPP-P-40 - Packaging and Packing of Hand Tools.

Federal Standards:

Fed. Std. No. 346 - Gages, Wrench Opening.
Fed. Std. No. 358 - Sampling Procedures.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Specifications:

MIL-L-7808 - Lubricating Oil, Aircraft Turbine Engine, Synthetic Base.
MIL-L-23699 - Lubricating Oil, Aircraft Turbine Engine, Synthetic Base.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

E18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Tests For.
D2240-68 - Standard Method of Test for Indentation Hardness of Rubber and Plastics by Means of a Durometer.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Stree, Philadelphia, PA 19103.)

American National Standards Institute (ANSI), Inc., Standards:

B107.4 - Driving and Spindle Ends for Portable Hand, Air and Electric Tools.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations shown herein are for the convenience of identification and are not intended to preclude the purchase of wrenches otherwise conforming to the requirements of this specification.

3.2 Materials. The sockets shall be manufactured of steel or powdered metal. The tool shanks shall be of solid or tubular steel. If of welded design, filler rod shall approximate the base metal in composition. The sockets and shanks shall be chromium plated unless otherwise specified. (see 6.1). Wrench handles shall be a transparent plastic (see 3.4).

3.3 Manufacture and design. Socket design shall be such that all applicable tests of section 4 herein shall be satisfactorily met. Steel sockets shall be through hardened to Rockwell C-35 to C-54 or case hardened 0.005 inch deep (min.) to a hardness of Rockwell 15N, 89 to 91. Powdered metal sockets shall be hardened to an average hardness of Rockwell C-30 to C-45. Average hardness shall be determined by arithmetic averaging of 3 test results taken in close proximity. The nut or engaging surfaces of the sockets shall be well defined. The socket opening dimensions and tolerances shall conform to the gaging requirements of Fed. Std. 346. Chromium plating shall be electrodeposited nickel with a minimum depth of 0.0002 inch followed by a 0.0001 inch coating of chrome. The socket and shank, after plating, shall have a maximum of 150 microinch (arithmetic average) finish. All edges, corners, etc. which may be potentially injurious shall be rounded, chamfered or otherwise smoothed. Each socket wrench shall be marked in a legible permanent manner with the nominal wrench opening size and a name or trademark such that the source of manufacture may be easily determined.

3.4 Plastic handles. Handles shall be transparent amber, or yellow in color, however, this does not preclude the furnishing of handles in accordance with the manufacturer's standard color coding practice. Handles shall be chemically polished or suitably finished to provide a comfortable grip. They shall be free from rough edges, sharp corners, or tool marks that adversely affect the appearance of the tool. Cross-sections shall be essentially square, hexagonal, triangular, round, or octagonal. Square, hexagonal, and triangular handles shall be rounded on the edges to a minimum radius equal to the distance across the flats divided by four. Round and octagonal handles shall be fluted with four to six major flutes for a length at least 65 percent of the handle length. The sides of the flute and the major periphery shall make an angle of not less than 90°, and the juncture shall be broken to 1/64 radius minimum. The handle butt shall be well rounded to a minimum radius to 1/8 of the major cross-sectional dimension and the run out of the radius with the major periphery shall not exceed an angle of 15°.

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3.4.1 Handle with female drive insert. When specified (see 6.1), handles shall have a metal square female drive insert of a design suitable for accepting a corresponding square male drive tang in accordance with table 9, ANSI B107.4. The insert shall be flush with the butt end of the screwdriver handle. If the insert is made of ferrous material, it shall be treated to prevent corrosion. The insert shall meet the test requirements in 4.3.2. When handles with inserts are furnished, the diameter of the handle shall not be less than 1-1/32 inch. One or more faces of the female drive opening shall be drilled or recessed so that it will engage the spring-loaded steel ball, or steel plunger on the corresponding male drive.

3.4.2 Cushion grip. When specified (see 6.1), handles shall be furnished with cushion grips. The grips shall be of rubber or vinyl capable of meeting the tests of 4.3.4. The length shall be at least 60 percent of the handle length, and there shall be no detectable slippage between the handle and the cushion grip under a normal usage test. The original hardness shall be Shore A 60 to 75 when tested in accordance with ASTM D2240-68, and the subsequent hardness after the brake fluid and oil test (see 4.3.4), shall not be greater than 80 Shore A. The cushioned handles shall meet the dimensional requirements for corresponding conventional type handles for each type, class, and style (see applicable tables) except that an additional 7/8 inch plus tolerance will be permitted on the handle length, and 3/16 inch plus tolerance will be permitted on the diameter of the handle. The fluting requirements for handles do not apply when cushion grips are supplied.

3.5 Type I, conventional, with solid or hollow shaft. The type I, wrenches shall consist of a round or hexagon steel shaft with a socket at one end and a plastic handle at the other end. The socket wrench opening shall be hexagon for sizes up to and including 11/32 inch and either hexagon or 12 point for 3/8 inch and larger sizes. A bolt clearance hole shall be provided in the wrench. The wrench shall be similiar to figure 1 and conform to table I for the size specified (see 6.1). "H" dimension on all sockets shall be 1 inch minimum.

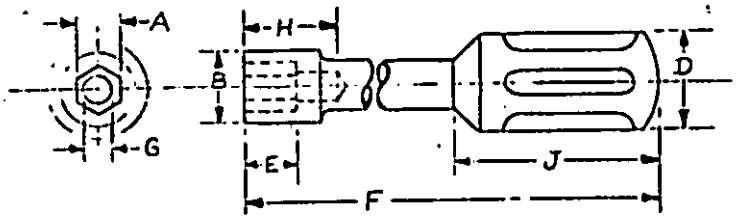


FIGURE 1. Type I, conventional with solid or hollow shaft.

TABLE I. Dimensions for type I, conventional size, with solid or hollow shaft

A	B	D	E	F	G	J	
Nominal socket opening	Socket diameter outside	Handle diameter outside	Depth of Broach	Overall length	Diameter of hole	Length of handle	Torsional test load
	Maximum	Minimum	Minimum	Minimum	Minimum	Minimum	Minimum
Inch	Inches	Inches	Inch	Inches	Inch	Inches	Inch-Pounds
5/32	0.335	7/8	3/16	6	0.100	3-1/8	50
3/16	.385	7/8	3/16	6	.105	3-1/8	75
7/32	.400	7/8	3/16	6	.115	3-1/8	75
1/4	.450	7/8	3/16	6	.118	3-1/8	100
9/32	.475	7/8	3/16	6	.129	3-1/8	100
5/16	.510	7/8	7/32	6	.141	3-1/8	125
11/32	.545	7/8	1/4	6	.170	3-1/8	125
3/8	.585	7/8	1/4	6	.196	3-1/8	150
7/16	.660	1	9/32	6-1/2	.265	3-1/2	150
1/2	.720	1	5/16	6-1/2	.328	3-1/2	175
17/32	.768	1-1/16	5/16	7	.391	3-5/8	175
9/16	.814	1-1/16	7/16	7	.391	3-5/8	175
5/8	.892	1-1/16	7/16	7	.445	3-5/8	200
11/16	.952	1-1/8	7/16	7	.445	3-5/8	250
3/4	1.054	1-1/8	7/16	7	.505	4	250
13/16	1.108	1-1/8	7/16	7	.567	4	250
7/8	1.201	1-1/8	7/16	7	.567	4	250

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3.6 Type II, pocket. The type II wrenches shall consist of a hexagonal or round steel shaft with a socket at one end and a plastic handle at the other end, with a bolt clearance hole in the shaft of the socket end. The wrench shall be similar to figure 2, shall conform to dimensions shown on figure 2, and conform to table II for sizes specified (see 6.1).

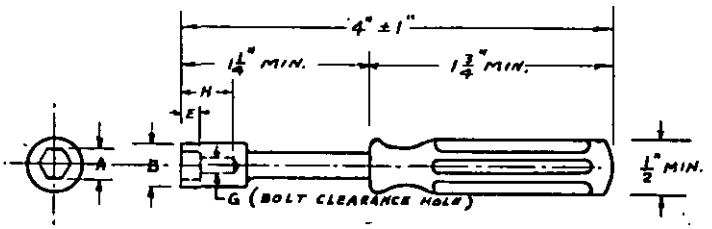


FIGURE 2. Type II, pocket.

TABLE II. Dimensions for type II, pocket

A	B	E	G	H	
Nominal socket opening	Socket dia. outside	Depth of broach	Dia. of hole	Depth of hole	Torsional test load
	Maximum	Minimum	Minimum	Minimum	Minimum
Inch	Inch	Inch	Inch	Inches	Inch-Pounds
5/32	0.335	5/32	0.100	1/2	30
3/16	.385	3/16	.105	1/2	45
7/32	.400	3/16	.115	1/2	45
1/4	.450	3/16	.118	1-1/16	60
9/32	.475	3/16	.129	1-1/16	60
5/16	.510	7/32	.141	1-1/16	75
11/32	.545	1/4	.170	1-1/16	75
3/8	.585	1/4	.196	1-1/16	90
7/16	.660	9/32	.265	1-1/16	100

3.7 Type III, miniature. Wrenches shall consist of a hexagonal or round steel shaft with either a hexagonal or 12-point socket at one end and a plastic handle at the other end, with a bolt clearance hole in the shaft of the socket end. The wrench shall be similar to figure 3, shall conform to dimensions shown on figure 3, and conform to table III for sizes specified (see 6.1).

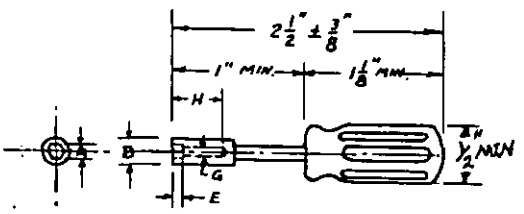


FIGURE 3. Type III, miniature.

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TABLE III. Dimensions for type III, miniature

A	B		E	C	H	
Nominal socket opening	Dia. or distance across corners	Socket opening		Depth of broach	Dia. of hole	Depth of hole	Torsional test load
	Maximum	Minimum	Maximum	Minimum	Minimum	Minimum	Minimum
Inch	Inch	Inch	Inch	Inch	Inch	Inch	Inch-Pounds
5/64	0.188	0.080	0.085	5/64	0.050	1/4	3
3/32	.188	.096	.101	3/32	.063	9/32	4
7/64	.204	.111	.116	7/64	.075	5/16	5
1/8	.234	.127	.132	1/8	.097	11/32	10
5/32	.266	.158	.163	5/32	.100	3/8	15

3.8 Type IV, stubby. Wrenches shall consist of a hexagonal or round steel shaft with either a hexagonal or 12-point socket at one end and a plastic handle at the other end, with a bolt clearance hole in the shaft of the socket end. The wrenches shall be similar to figure 4, shall conform to the dimensions shown on figure 4, and shall conform to table IV for sizes specified (see 6.1).

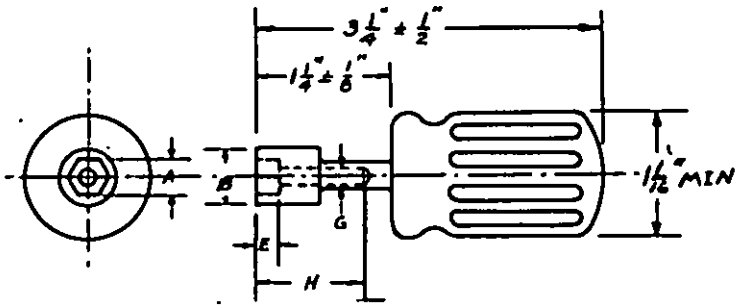


FIGURE 4. Type IV, stubby.

TABLE IV. Dimensions for type IV, stubby

A	B	E	C	H	
Nominal socket opening	Dia. or distance across corners	Depth of broach	Dia. of hole	Depth of hole	Torsional test load
	Maximum	Minimum	Minimum	Minimum	Minimum
Inch	Inch	Inch	Inch	Inches	Inch-Pounds
1/4	0.450	3/16	0.118	1-1/16	100
5/16	.510	7/32	.141	1-1/16	125
11/32	.545	1/4	.170	1-1/16	125
3/8	.585	1/4	.196	1-1/16	150

3.9 Sets. When sets of socket wrenches are required, the size of the tools for the particular set should be specified in the contract or order (see 6.1). The components of the sets shall conform to the requirements of the respective items in this specification.

3.9.1 Wrap or rolls. Spin type socket wrenches, when ordered in sets, shall be furnished in durable fabric or plastic wraps or rolls. Sets furnished in fabric or plastic wraps or rolls shall contain compartments for each tool in the set and the wrap or roll shall be furnished with durable ties.

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3.10 Workmanship. The wrenches shall be free from rust, fins, burrs, sharp edges, and other defects which may impair their serviceability, durability or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with FED-STD-358.

4.3 Testing. Each sample unit shall be tested in accordance with 4.3.1 and 4.3.2. Additional tests shall be as described in 4.3.3.

4.3.1 Hardness. Hardness tests shall be conducted in accordance with ASTM E-18. When grinding is necessary to prepare the test surface, the amount of material removed must not exceed 0.007 inches on the surface contacted by the indentor.

4.3.2 Load. Load tests shall be conducted by using a hexagonal test mandrel as described in table V, hardened to Rockwell C55 minimum, affixed to a torque measuring device. The tests shall be conducted by preheating the tool to a uniform temperature of 125°F and the load applied within 1 minute after removing the tool from the heating medium. The hexagonal socket of the tool shall be inserted over the mandrel to the depth prescribed in table V. The load shall then be applied by twisting the plastic handle and sustaining the required load for a minimum of 10 seconds. The handle may be gripped on the end most extreme from the socket by means of a pair of pliers or the equivalent. Any cracking, excessive brinelling, slippage of the handle with respect to the shank, or failure to sustain the test load shall constitute rejection of the tool.

TABLE V. Hexagonal mandrel dimensions and maximum depth of mandrel insertion (inches)

Nominal size of wrench opening	Hexagonal mandrel dimensions		Maximum depth of mandrel insertion
	Across flats tolerances	Across corners minimum	
5/32	+.001 -.002	.1745	.094
3/16	+.001 -.002	.2095	.094
7/32	+.001 -.002	.2440	.109
1/4	+.001 -.002	.2780	.125
9/32	+.001 -.002	.3133	.141
5/16	+.001 -.002	.3495	.141
11/32	+.001 -.002	.3860	.156
3/8	+.001 -.002	.4225	.156
7/16	+.001 -.002	.4935	.218
1/2	+.001 -.003	.5635	.265
9/16	+.001 -.003	.6339	.328
5/8	+.001 -.003	.7055	.375
11/16	+.001 -.003	.7769	.375
3/4	+.001 -.003	.8485	.437
13/16	+.001 -.003	.9201	.453
7/8	+.001 -.003	.9917	.500

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4.3.3 Plastic handle test. Two sample wrenches shall be selected from the sample lot representing all wrenches having plastic handles or handles with cushion grips. Each sample shall resist the following tests without damage; immerse the handle of first sample in brake fluid conforming to VV-B-680, remove and let stand for 72 hrs. and test to applicable torque load. Repeat the test (using the second sample wrench) with oil conforming to either MIL-L-7808 or MIL-L-23699. If the wrench has a cushion grip, the change in the grip material shall be such that the hardness is no greater than 80 Shore A. Failure of either of the sample wrenches shall constitute failure of the lot.

4.4 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements of PPP-P-40.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing and marking shall be in accordance with PPP-P-40. Preservation, packaging and packing shall be level A, B, or C, as specified (see 6.1).

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and size of tool required (see 1.2 and applicable tables).
(c) If cushion grips are required for type I wrenches, so state (see 3.4.2).
(d) Handles with metallic inserts (see 3.4.1).
(e) The number of parts comprising a set, and the size of the sockets when sets are required (see 3.9).
(f) Applicable levels of preservation, packaging, and packing (see section 5).

MILITARY INTEREST:

User activities:

Army - GL, AV
Navy - SH

Preparing activities:

GSA-FSS

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